Session 6
Lab Report
Dice Roller

Name : Pawankumar

Course : Android App Development with Gen AI

Dice Roller Program using Kotlin

Abstract

This report explains the implementation of a Dice Roller program using Kotlin. The application simulates rolling two dice using random number generation. When both dice show the same number, the program displays a special message to make the output more interactive.

Objective

- Understand Kotlin random number generation.
- Implement modular functions in Kotlin.
- Simulate dice rolling using programming logic.
- Display special output when both dice values match.

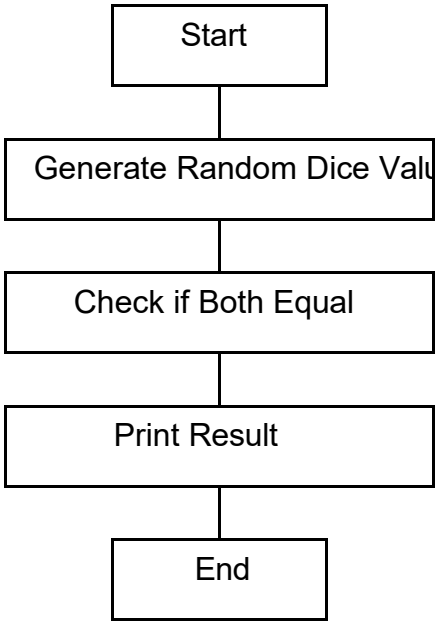
Introduction

Dice simulation is a common beginner programming exercise. Using Kotlin, we can easily simulate dice rolling by generating random numbers between 1 and 6. This project demonstrates how programming can replicate real-world events like dice rolling while also practicing functions, conditionals, and program structure.

Algorithm

1. Start the program.
2. Create a random number generator.
3. Roll the first die using a random number from 1 to 6.
4. Roll the second die using a random number from 1 to 6.
5. Print both dice values.
6. Check if both values are equal.
7. If equal, display 'Double!'.
8. Repeat for multiple rolls.
9. End the program.

Flowchart

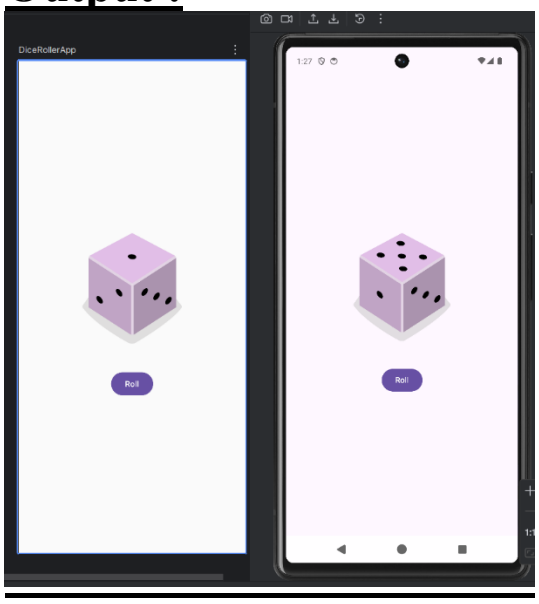


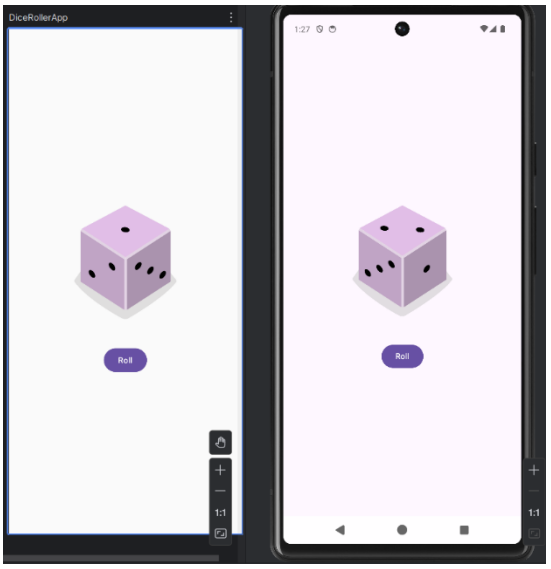
Kotlin Source Code

```
package com.example.diceroller

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.Image
import androidx.compose.foundation.layout.*
import androidx.compose.material3.Button
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Surface
import androidx.compose.material3.Text
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.res.painterResource
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp

class MainActivity : ComponentActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      enableEdgeToEdge()
      setContent {
         Surface(
            modifier = Modifier.fillMaxSize(),
            color = MaterialTheme.colorScheme.background
         ) {
            DiceRollerApp()
         }
      }
   }
}

@Composable
fun DiceWithButtonAndImage(modifier: Modifier = Modifier) {

   var result by remember { mutableStateOf(1) }
```

```
 val imageResource = when (result) {
      1 -> R.drawable.dice_1
      2 -> R.drawable.dice_2
      3 -> R.drawable.dice_3
      4 -> R.drawable.dice_4
      5 -> R.drawable.dice_5

 else -> R.drawable.dice_6
  }

  Column(
      modifier = modifier,
      horizontalAlignment = Alignment.CenterHorizontally,
      verticalArrangement = Arrangement.Center
  ) {

      Image(
          painter = painterResource(id = imageResource),
          contentDescription = result.toString()
      )

      Spacer(modifier = Modifier.height(16.dp))

      Button(
          onClick = { result = (1..6).random() }
      ) {
          Text(text = "Roll")
      }
  }
}

@Preview(showBackground = true)
@Composable
fun DiceRollerApp() {
   DiceWithButtonAndImage(
      modifier = Modifier
          .fillMaxSize()
          .wrapContentSize(Alignment.Center)
   )
}
```

Output :





Conclusion

The Dice Roller Kotlin program demonstrates how random number generation and functions can simulate real-world actions like dice rolling. The project strengthens understanding of program logic, modular programming, and Kotlin syntax.

References

1. Kotlin Official Documentation (kotlinlang.org)
2. Android Developers Kotlin Guide
3. Kotlin Programming Books and Tutorials